

Why We Use querystring Module in Node.js

1. JSON.parse vs querystring.parse

JSON.parse() is used to convert JSON formatted string into an object, while the querystring module is used to convert URL-encoded data (query strings) into an object.

2. Query String Example (NOT JSON Format)

Example of a query string:

name=Malaika&age;=21&city;=Lahore

This is NOT JSON, so JSON.parse() cannot read it.

3. JSON Format Example

```
{ "name": "Malaika", "age": 21, "city": "Lahore" }
```

This can be parsed using JSON.parse().

4. Why JSON.parse() Cannot Handle Query Strings

If you try JSON.parse("name=Malaika&age;=21"), it will throw an error because the string is NOT valid JSON.

5. What querystring.parse() Does

querystring.parse() converts URL query string data into a JavaScript object. For example:

Input: name=Malaika&age;=21

Output: { name: 'Malaika', age: '21' }

6. When Query Strings Are Used

Query strings are used when data comes from:

- URL parameters
- GET forms
- Browser address bar
- Redirect links

7. Summary (Simple One Line)

We use `querystring` because URLs use a different format than JSON, and `JSON.parse` cannot read URL-encoded data.